Md. Shakil Ahmed

Jahangirnagar University, Savar, Dhaka-1342, Bangladesh
shakil.ahmed@bubt.edu.bd Website Google Scholar GitHub

RESEARCH INTERESTS

Cyber Security, Federated Learning, Privacy-Preserving and Trustworthy AI, Large Language Model, Machine Learning, Deep Learning, Healthcare AI, Computer Vision

EDUCATION

Jahangirnagar University, Savar, Dhaka, Bangladesh **Sep 2024 – Sep 2026**
M.Sc., Information and Communication Technology ***CGPA:*** *3.81/4.00 (4th position)*

Jahangirnagar University, Savar, Dhaka, Bangladesh **Feb 2019 – Sep 2024**
B.Sc., Information and Communication Technology ***CGPA:*** *3.86/4.00 (4th position)*

EXPERIENCES

Lecturer, Department of Computer Science and Engineering **Jul, 2025 - Present**
Bangladesh University of Business and Technology (BUBT)

Lecturer, Department of Computer Science and Engineering **Mar, 2025 - Jul, 2025**
Canadian University Of Bangladesh (CUB)

Lecturer, Department of Software Engineering **Jan, 2025 - Feb, 2025**
Daffodil International University (DIU)

Lecturer, Department of Computer Science and Engineering **Sep, 2024 - Dec, 2024**
Daffodil International University (DIU)

Teaching Assistant, Institute of Information Technology (IIT) **May, 2024 - Nov, 2024**
Jahangirnagar University (JU)

Research Assistant, Institute of Information Technology (IIT) **Jan, 2023 - Jul, 2024**
Jahangirnagar University (JU)

HONORS AND AWARDS

Information and Communication Technology (ICT) Division Fellowship **2025 – 2026**
Title: *FEP-SSL:A Privacy-Preserving Semi-Supervised Learning Framework for Parkinson's Disease Classification.*

National Science and Technology (NST) Fellowship **2025 – 2026**
Title: *Privacy Preserving Federated Semi-Supervised Learning Model for Healthcare Applications.*

Govt. Scholarship, University Merit Scholarship for Academic Results of B.Sc. **2019 – 2024**

RESEARCH PUBLICATIONS

Journal Papers

1. Ahmed, K. R., Ali, M. S., Rahman, A., Sharmin, S., **Ahmed, M. S.**, Al Farid, F., Mansor, S., and Miah, A. S. M. (2026). "Enhanced Detection of GAN-Generated Image Forgeries Using Error Level Analysis and Deep Learning Models." *IET Image Processing*, *20*(1), p. e70345. DOI: 10.1049/ipr2.70345.
2. **Under Review: Ahmed, M. S.**, Shafiq, S., Kaiser, M. S., Rahman, M. S., Hosen, A. S. M., and Hwang, Y., "FEP-SSL: A Privacy-Preserving Semi-Supervised Learning Framework for Parkinson's Disease Classification," *PLOS ONE*, 2026.
3. **Under Review:** "An Explainable Ensemble Machine Learning Framework for Early Prediction of Student Dropout Risk in Higher Education," *Scientific Reports*, 2026.
4. **Under Review:** "HDCNN: A Fairness-Aware Hybrid Deep Learning Framework for Skin Cancer Classification," *Springer Nature*, 2026.

Conference Papers

1. **Accepted: Ahmed, M. S.**, Shafiq, S., and Kaiser, M. S. (2026, August). "An Explainable Cleanlab-Guided Uncertainty-Based Active Learning Framework for Data-Efficient Parkinson's Disease Classification." In *6th International Conference on Applied Intelligence and Informatics (AII 2026)*. Springer.

2. **Accepted: Ahmed, M. S.**, Shahid, S. B., and Rashid, M. M. O. (2026). "Data Transmission Technologies: A Review of Security, Reliability, Efficiency, and Emerging Intelligent Communication." In *11th IEEE CSDE 2026, Asia-Pacific Conference on Computer Science and Data Engineering (IEEE CSDE 2026).*

3. **Accepted: Ahmed, M. S.**, Shahid, S. B., and Rashid, M. M. O. (2026). "Toward Intelligent Cauliflower Disease Recognition: A CLAHE-Enhanced Image Processing and SVM-Based Classification Framework." In *11th IEEE Asia-Pacific Conference on Computer Science and Data Engineering (IEEE CSDE 2026).*

4. **Accepted: Ahmed, M. S.**, Ahmad, F., Shahid, S. B., and Rashid, M. M. O. (2026). "A Comprehensive Structural Analysis of Zero-Day Attack Detection: Machine Learning Paradigms, Performance Trends, and Open Research Challenges." In *11th IEEE Asia-Pacific Conference on Computer Science and Data Engineering (IEEE CSDE 2026).*

5. Sohana, R. R., **Ahmed, M. S.**, Shahid, S. B., and Hridoy, R. H. (2025). "Robust Blood Cell Detection and Counting Using Efficient Channel Attention-Integrated YOLOv10." In *2025 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)* (pp. 1090–1095). IEEE. DOI: 10.1109/WIECON-ECE69386.2025.11525973.

6. Hosen, M. F., Khatun, A., Ali, M. S., **Ahmed, M. S.**, Sharmin, S., and Uddin, M. S. (2026). "Gastric Cancer Classification Using a Hybrid CNN-Attention-LSTM Model with Deep Feature Extraction." In *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)* (pp. 1–6). IEEE. DOI: 10.1109/QPAIN69676.2026.11546573.

7. Khatun, A., Hosen, M. F., Ali, M. S., Hasan, T., **Ahmed, M. S.**, and Sharmin, S. (2026). "Candidate Biomarkers and Key Molecular Pathways in Anxiety Disorder and Parkinson's Disease Using Bioinformatics and Machine Learning." In *2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII)* (pp. 1–6). IEEE. DOI: 10.1109/PECCII70991.2026.11662103.

Book Chapter

1. **Ahmed, M.S.**, Hossain, M., Amit, A. A., Tahsin, M., Mahmud, M., and Kaiser, M. S. (2023, November). "Real-Time Bangla Sign Language Detection and Recognition Using YOLOv10." In *International Conference on Trends in Computational and Cognitive Engineering* (pp. 383–400). Springer Nature Singapore. DOI: 10.1007/978-981-96-0185-1_33.

2. **Accepted:** Ahmad, F., Podder, S., Karmokar, S., Hossain, M. S., Rashid, M. M. O., **Ahmed, M. S.***, and Shahid, S. B., "A Multi-Lingual Cyberbullying Detection and Blockchain-Integrated Privacy-Preserving Federated Learning Framework," In 6th International Conference on Applied Intelligence and Informatics (AII 2026). Springer.

DATASET PUBLICATIONS

Dataset Publications

1. Amit, A. A., Hossain, M.M, **Ahmed, M. S.** Kaiser, M. S. (2025). "Real-Time Detection of Bangla Sign Language for Shopkeeper-Customer Communication," *Mendeley Data*, V1. DOI: 10.17632/kyb87r9x2w.1.

ACADEMIC PROJECTS

Learning Management System | *React, Node.js, Express, MongoDB* **GitHub Link**

- A role-based learning management system that enables instructors to schedule and manage course content releases according to diverse teaching strategies. The platform also facilitates collaborative discussions and real-time communication among users.

Ecommerce Website | *HTML, CSS, React, JavaScript, Express.js, Node.js, MongoDB* **GitHub Link**

- A role-based university cafeteria platform for online food ordering, with dedicated modules for managing menus, users, orders, payments, inventory, and deliveries.

QR Code Scanner | *Java (Android)* **GitHub Link**

- An Android application for generating QR codes from text and scanning QR codes to decode and display the encoded information.

Remote Control Fire Detection Car | *Arduino UNO, Sensors* **GitHub Link**

- An Arduino-based firefighting vehicle designed to detect fire, navigate toward the affected area, and spray water for fire suppression.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, PHP, Latex, HTML, CSS, JavaScript, Mysql
Machine Learning: PyTorch, Keras, OpenCV, TensorFlow, NLTK, Pandas, NumPy, Matplotlib
Developer Tools: VS Code, CodeBlocks, LATEX, PyCharm, Jupyter Notebook, Eclipse
Illustration Tools: Illustrator, Photoshop, Canva, Animate, Premier Pro

TRAINING

Cyber Security, Enhancing Digital Government and Economy (EDGE) — Sep 2024 – Dec 2024
Software Testing, Enhancing Digital Government and Economy (EDGE) — Mar 2024 – May 2024
Youth Social Leadership, Bangladesh Youth Leadership Training (BYLT) — Jan 2020 – Mar 2020
Graphics Design, Learning & Earning Development Project (LEDP) — April 2020 – Sep 2020

CO-CURRICULAR ACTIVITIES AND SERVICES

Admission Helpline Assistant, Jahangirnagar University Admission — 2023 – 2024
General Secretary, Dinajpur Students Association of Jahangirnagar University — 2022 – 2025
General Member, Jahangirnagar University Science Club — 2019 – 2022
Volunteer, Savar Half Marathon — 2022